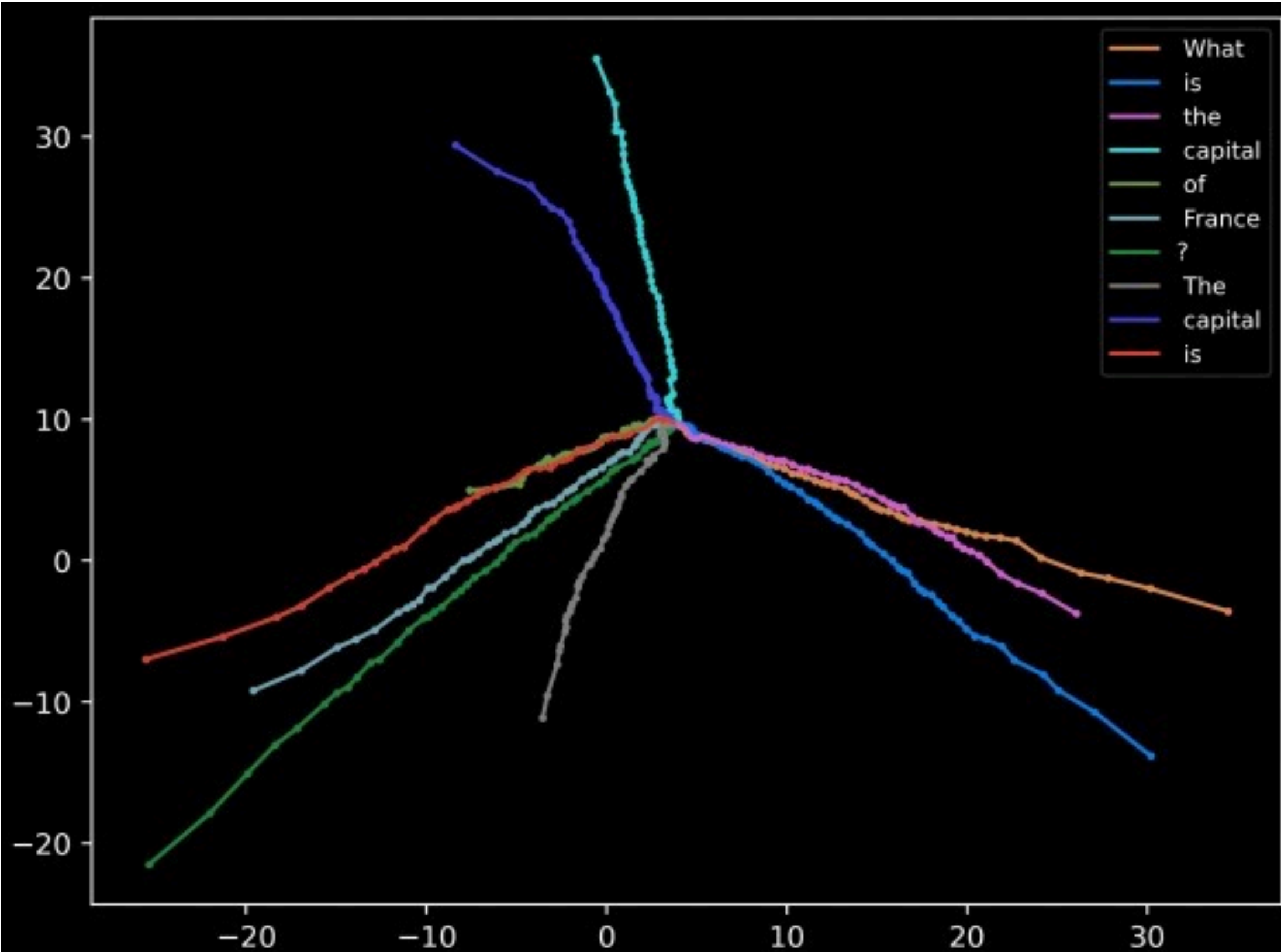


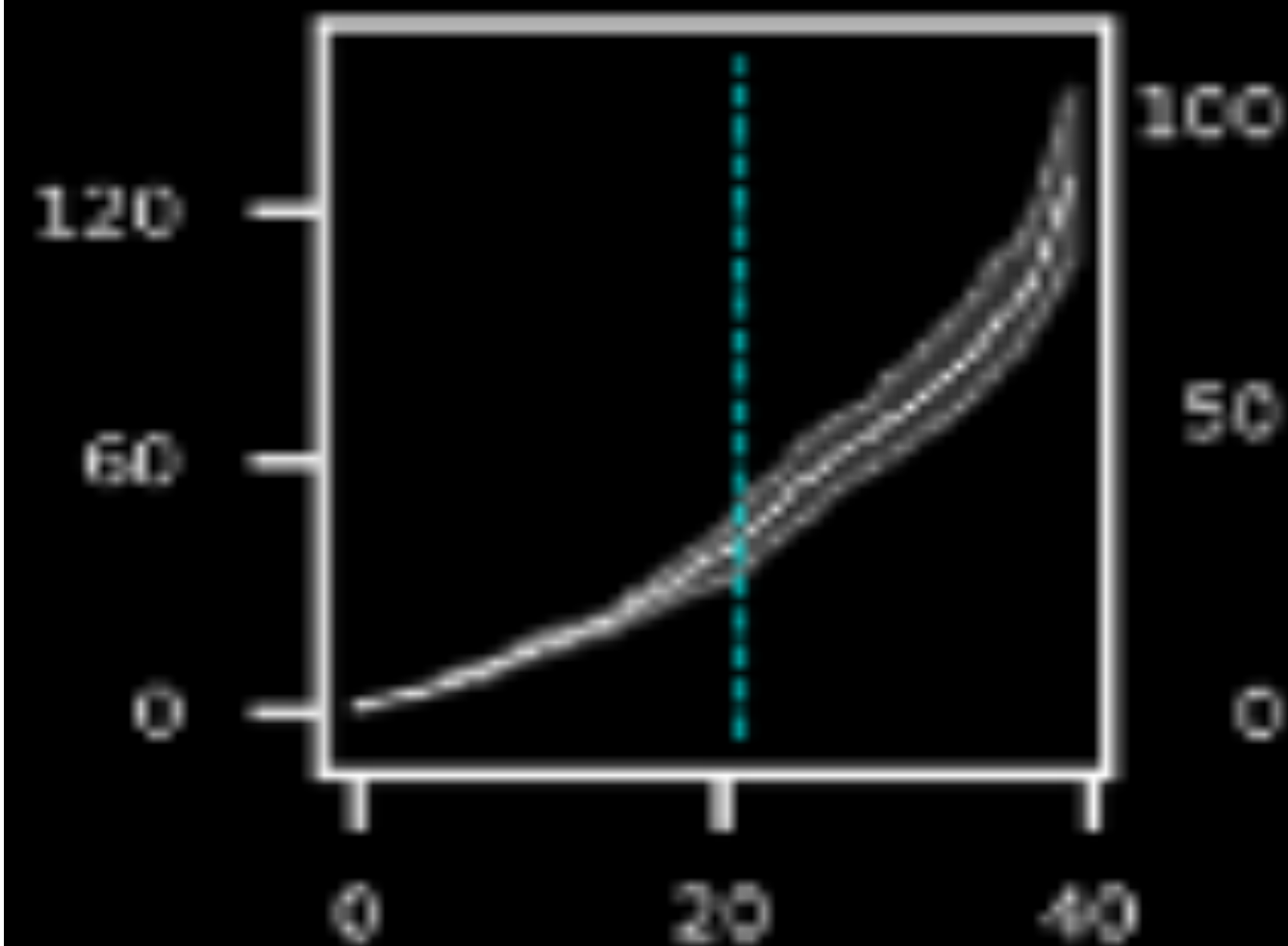


(a) Llama-3 70B

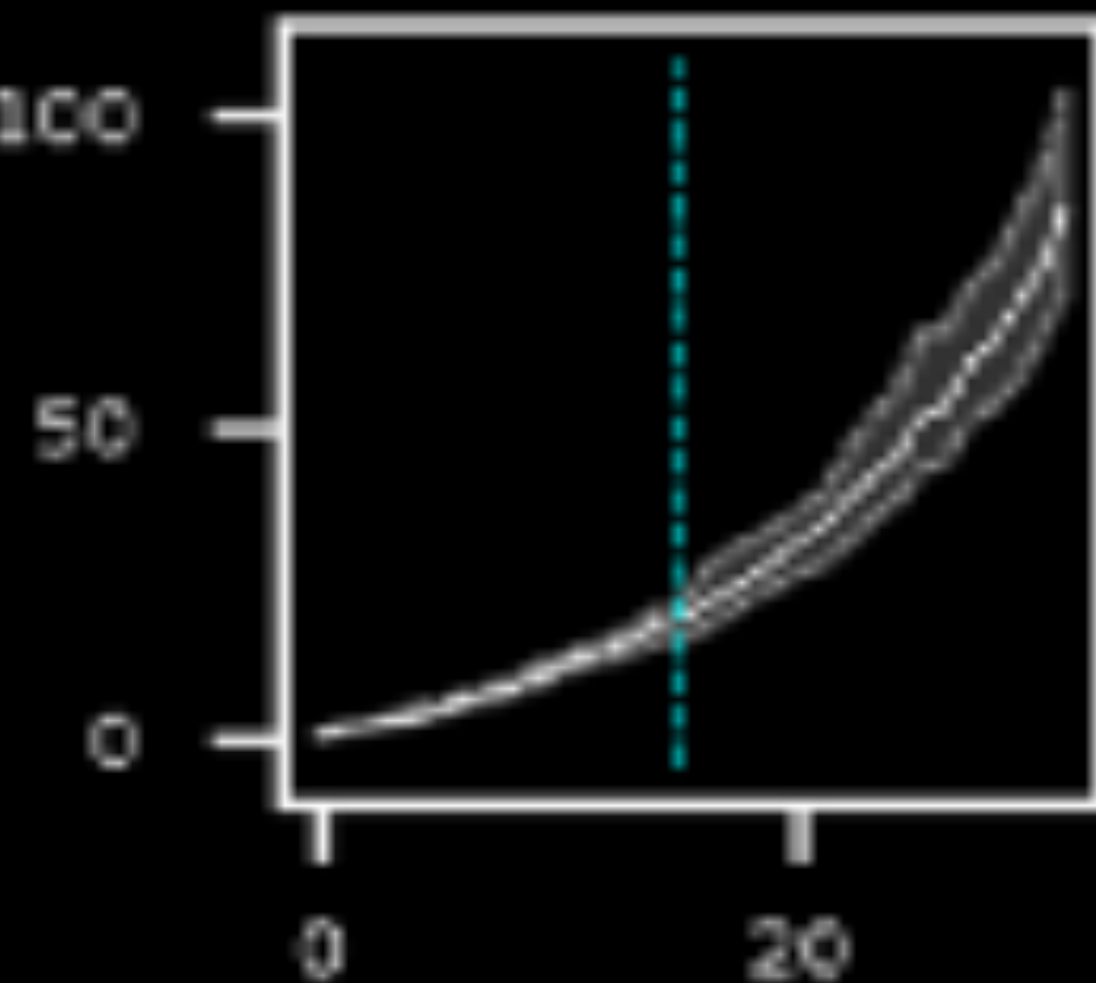


Representation Norms

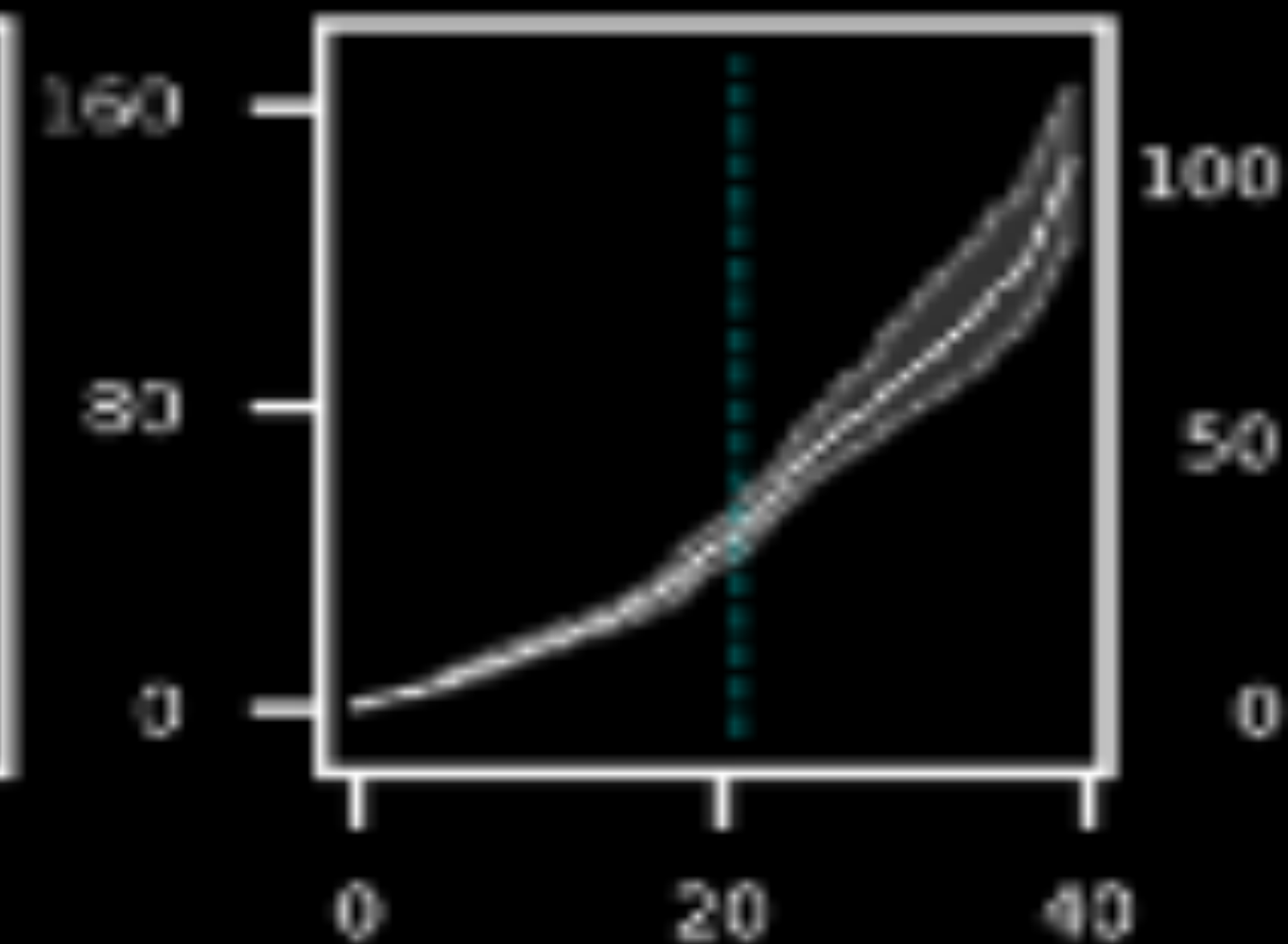
Orca 13B



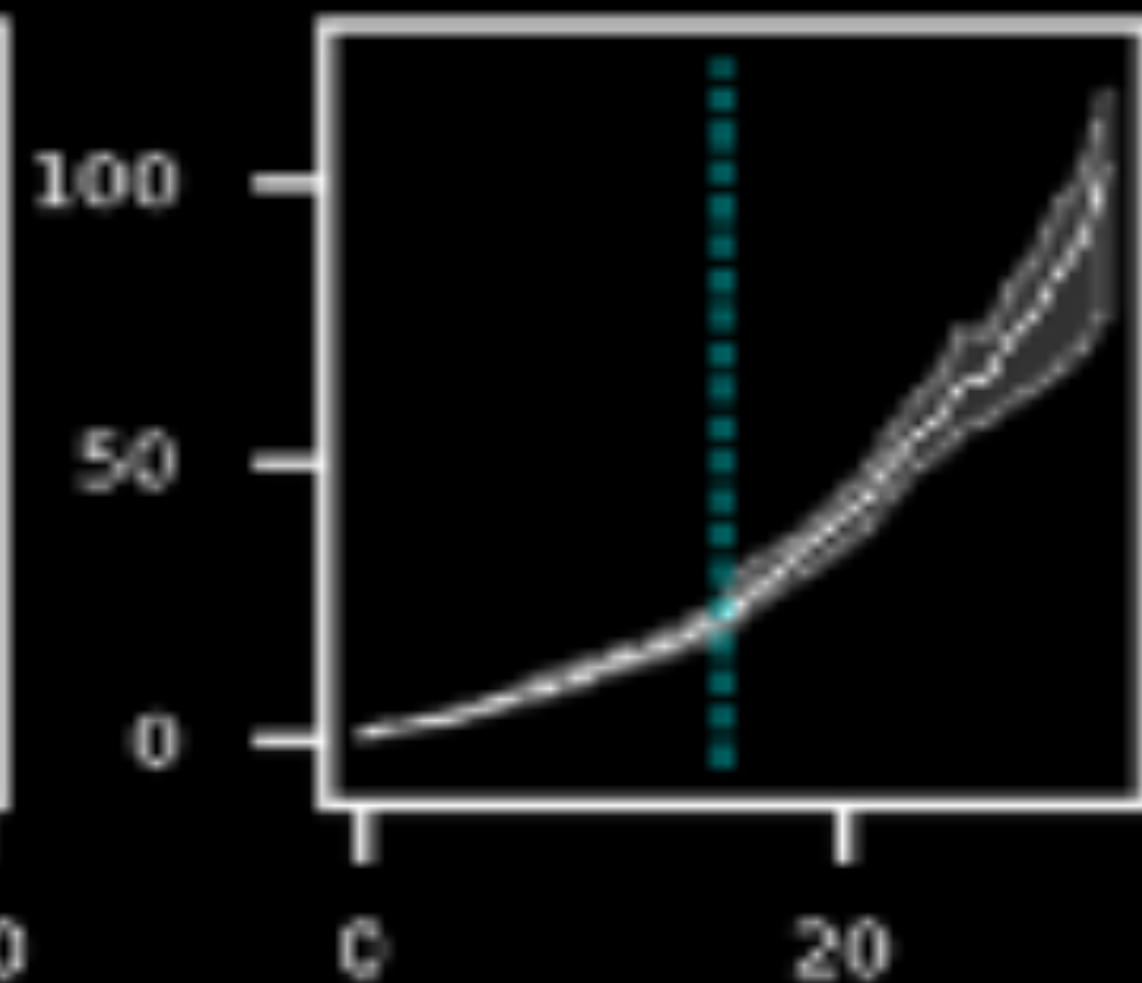
Orca 7B



Llama2 13B

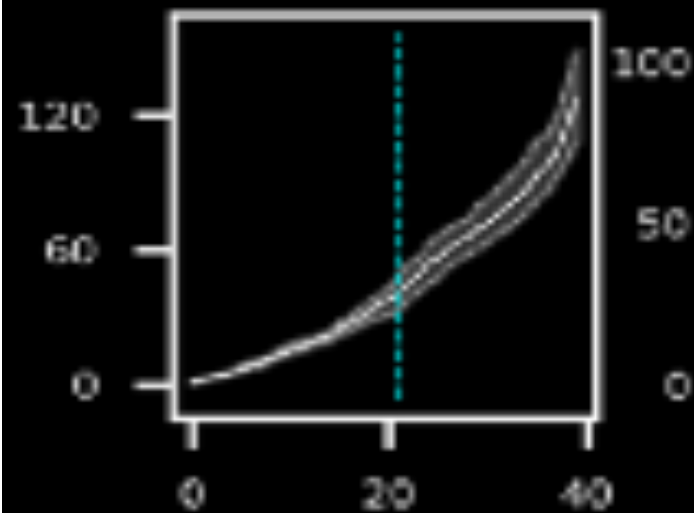


Llama2 7B

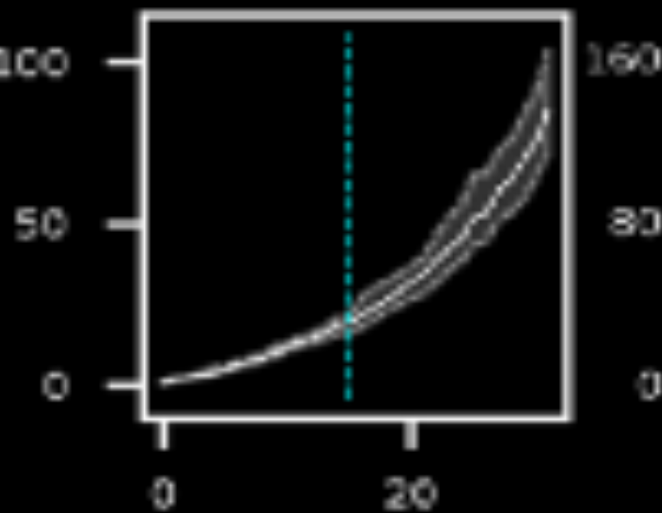


Representation Norms

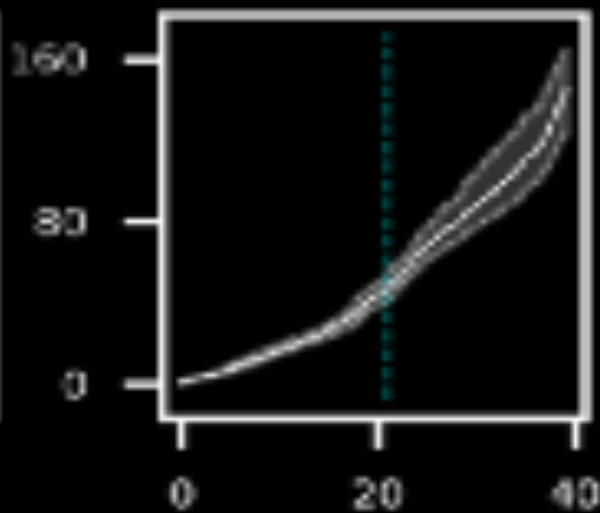
Orca 13B



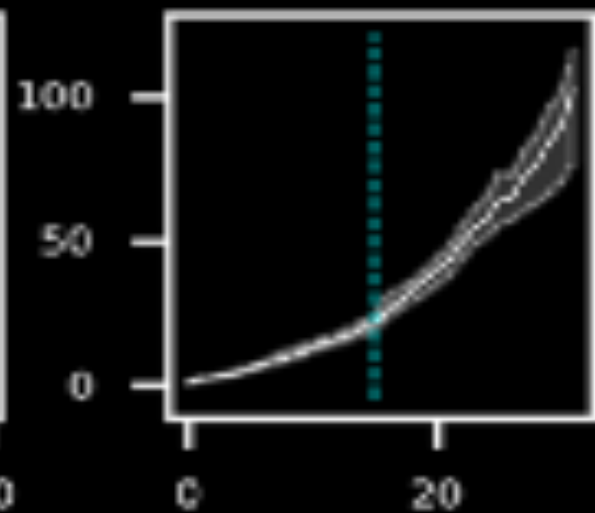
Orca 7B



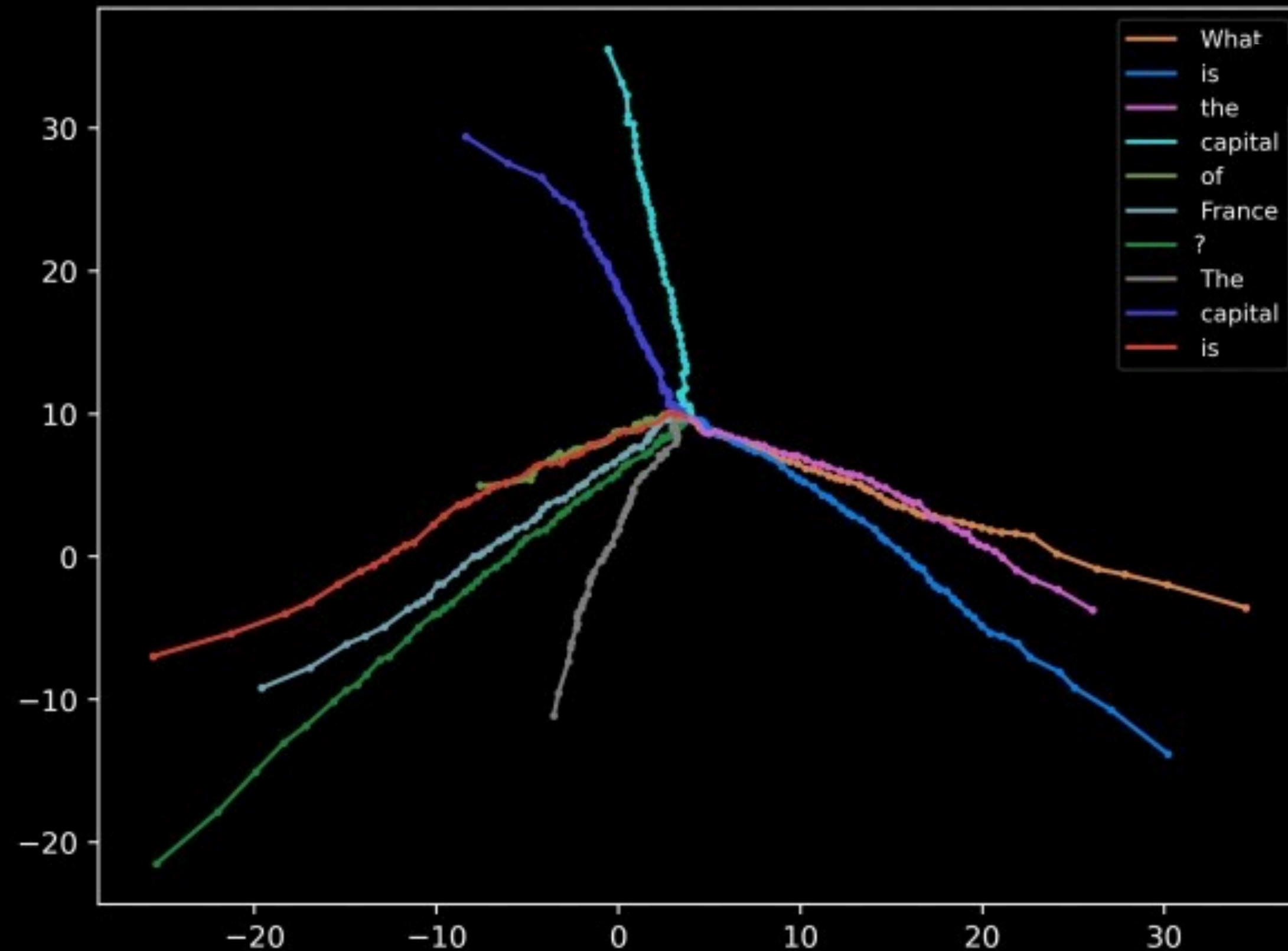
Llama2 13B



Llama2 7B

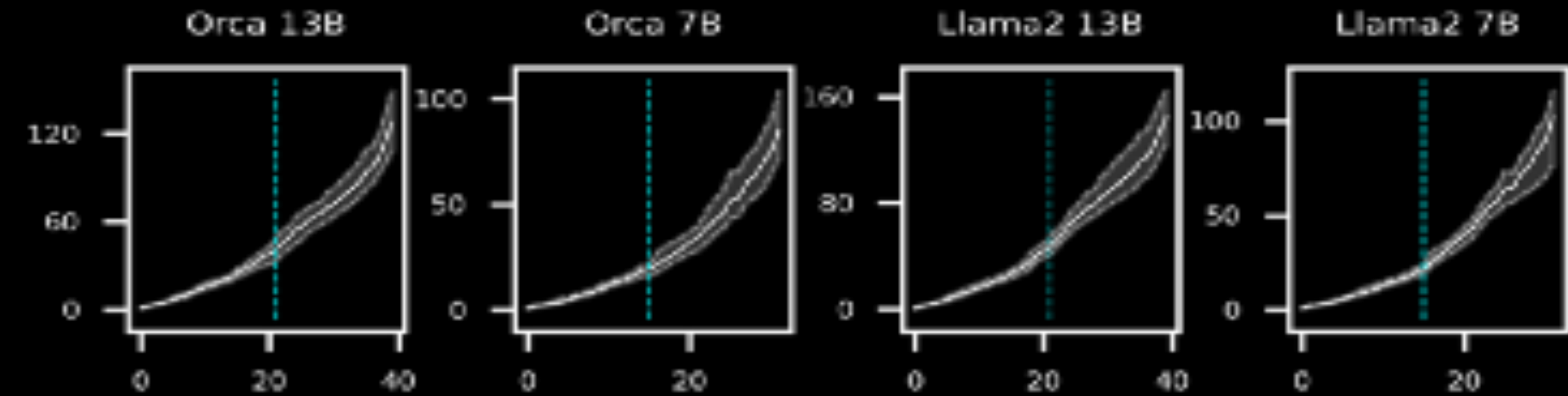


Representation Trajectories



(a) Llama-3 70B

Representation Norms



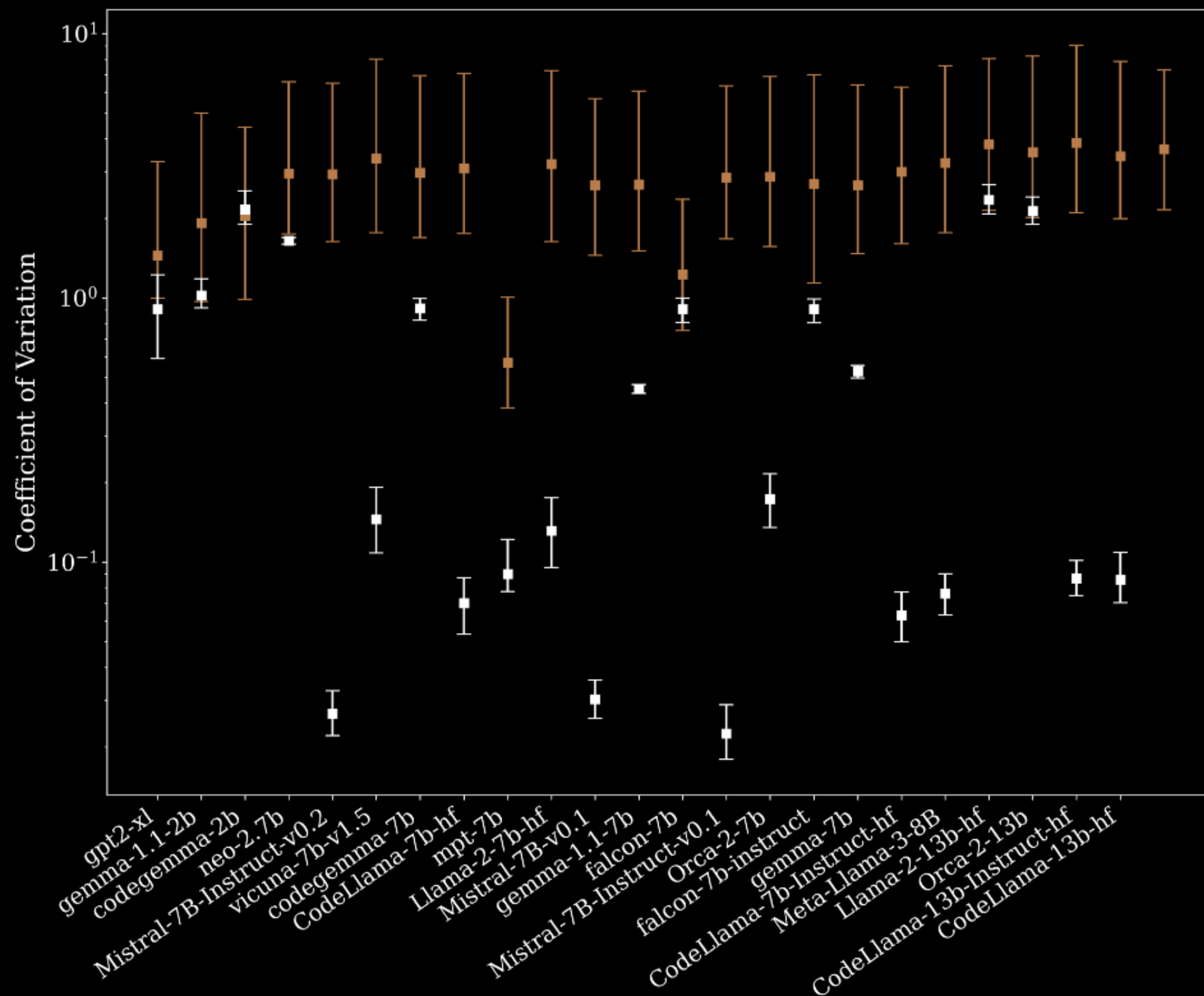
$$\|x^\ell\| \stackrel{?}{=} \|x^0\| \exp(\alpha_\ell \ell)$$

$$\alpha_\ell \approx \frac{1}{\ell} \ln \left(\frac{\|x^\ell\|}{\|x^0\|} \right)$$

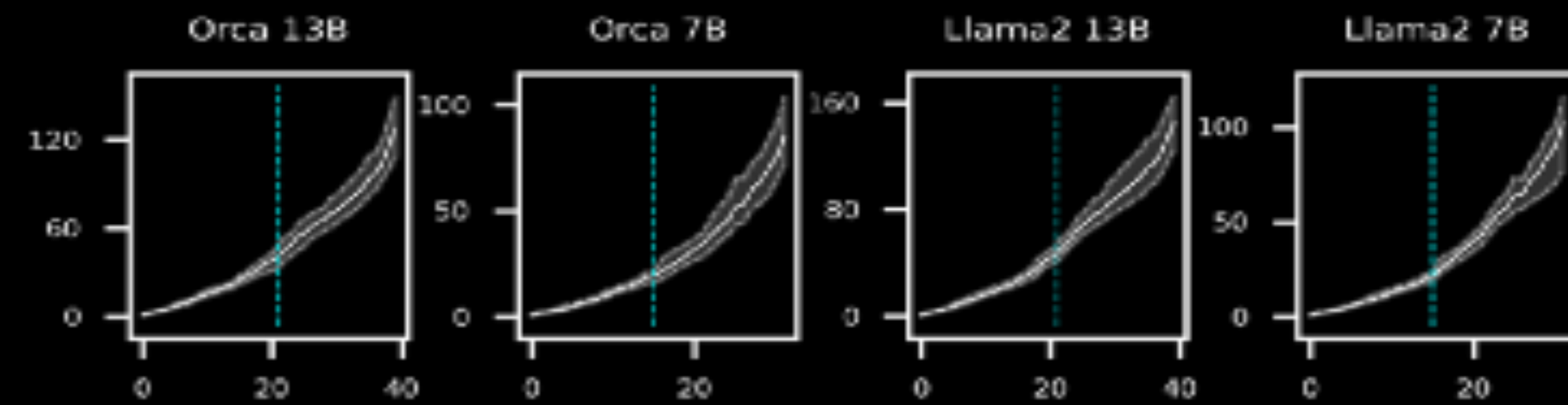
$$\text{ED} = \frac{\text{Var}_l \alpha^l}{(\text{Avg}_l \alpha^l)^2}$$

Representation Trajectories

Exponential Distance



Representation Norms



$$\|x^\ell\| \stackrel{?}{=} \|x^0\| \exp(\alpha_\ell \ell)$$